Alarm Copilot

Operations intelligence · Local Ollama

Copilot backend connected

8 MCP tools discovered

Investigation sources

- [x] Live alarm data (MCP)
- [x] Operating procedures (RAG)

Disable either source to run a focused investigation.

Try an investigation

Show active critical alarms for Boiler Feed Pump 101 and recommend immediate actions

Why are compressor discharge pressure alarms repeatedly occurring? What's the pattern?

Which alarm has the highest priority in EastRefinery, and what should be done?

What related assets should be inspected for this motor trip alarm? What's the operating procedure?

Clear conversation

Alarm investigation workspace

Investigate alarms using live system data and approved procedures. Results include source citations and the MCP execution trace.

What would you like to investigate?

Ask about an alarm, asset, or recurring pattern. The copilot can retrieve current alarm context and relevant opera

- Resolve asset and alarm context
- Retrieve procedures and guidance
- Review evidence and tool trace

Show active critical alarms for Boiler Feed Pump 101 and recommend immediate actions

The provided data appears to be a JSON-formatted log file containing alarm data for a boiler feed pump sy alarms with different alarm IDs, asset IDs, alarm names, severities, and statuses. Here is a summary of the

Alarms:

1. **ALM-A001-003**: Flow Rate Low (Severity: Critical, Status: Active)
   - Start Time: 2026-08-31T18:32:32.777685Z
   - Asset Name: Boiler Feed Pump 101
2. **ALM-A001-002**: Temperature High (Severity: High, Status: Acknowledged)
   - Start Time: 2026-08-25T18:32:32.777685Z
   - Asset Name: Boiler Feed Pump 101
3. **ALM-A001-001**: Vibration Alarm (Severity: Medium, Status: Cleared)
   - Start Time: 2026-08-19T18:32:32.777685Z
   - Asset Name: Boiler Feed Pump 101
4. **ALM-A001-004**: Motor Trip (Severity: Low, Status: Cleared)
   - Start Time: 2026-09-06T18:32:32.777685Z
   - Asset Name: Boiler Feed Pump 101
5. **ALM-A001-005**: High Discharge Pressure (Severity: Medium, Status: Acknowledged)
   - Start Time: 2026-09-12T18:32:32.777685Z
   - Asset Name: Boiler Feed Pump 101

Recommendations:

1. **ALM-A001-003**:
   - Immediate Actions: Verify the Flow Rate Low indication against current instrument readings; Follo operating limits before taking action.
   - Preventive Measures: Schedule preventive maintenance; Review operating procedures.
   - Related Assets: A002, A003
   - Documentation References: SOP-101, Maintenance-Manual-Ch3
   - Trace ID: acd895c4-93b9-44c2-9249-fb6537651074
   - X-Client ID: alarm-copilot

Overall, the log file indicates that several alarms have been triggered, including Flow Rate Low, Temperatu High Discharge Pressure. The system has taken actions to address some of these

Procedure sources	MCP tool calls	Evidenc
4	3	80%

Latency — total 172521 ms · model 170264 ms across 1 call(s) · MCP tool-time sum 733 ms · RAG 179 ms · MCP connect/dis

Alarm and asset summary

Search Assets


```
{
  "total" : 1
  "results" : [
    0 : {
      "asset_id" : "A001"
      "asset_name" : "Boiler Feed Pump 101"
      "asset_type" : "Pump"
      "unit" : "Unit 1"
      "location" : "EastRefinery"
      "criticality" : "high"
    }
  ]
}
```


Get Alarms


```
{
  "total" : 15
  "page" : 1
  "page_size" : 200
  "total_pages" : 1
  "data" : [
    0 : {
      "alarm_id" : "ALM-A001-014"
      "asset_id" : "A001"
      "alarm_name" : "Motor Trip"
      "severity" : "high"
      "status" : "acknowledged"
      "start_time" : "2026-11-05T18:32:32.777685Z"
      "asset_name" : "Boiler Feed Pump 101"
      "alarm_type" : "safety"
      "description" : "Motor protection trip activated"
      "end_time" : NULL
      "ack_time" : "2026-11-05T19:02:32.777685Z"
      "ack_user" : "operator-1"
      "count" : 3
      "recurring" : true
    }
    1 : {
      "alarm_id" : "ALM-A001-013"
      "asset_id" : "A001"
      "alarm_name" : "Flow Rate Low"
      "severity" : "medium"
      "status" : "cleared"
      "start_time" : "2026-10-30T18:32:32.777685Z"
      "asset_name" : "Boiler Feed Pump 101"
      "alarm_type" : "process"
      "description" : "Flow rate below minimum threshold"
```

```
    "end_time" : "2026-10-30T20:32:32.777685Z"
    "ack_time" : NULL
    "ack_user" : NULL
    "count" : 3
    "recurring" : true
  }
  2 : {
    "alarm_id" : "ALM-A001-012"
    "asset_id" : "A001"
    "alarm_name" : "Temperature High"
    "severity" : "low"
    "status" : "active"
    "start_time" : "2026-10-24T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Operating temperature above safe limit"
    "end_time" : NULL
    "ack_time" : "2026-10-24T19:02:32.777685Z"
    "ack_user" : "operator-1"
    "count" : 3
    "recurring" : true
  }
  3 : {
    "alarm_id" : "ALM-A001-011"
    "asset_id" : "A001"
    "alarm_name" : "Vibration Alarm"
    "severity" : "critical"
    "status" : "acknowledged"
    "start_time" : "2026-10-18T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "device"
    "description" : "Excessive vibration detected"
    "end_time" : NULL
    "ack_time" : NULL
    "ack_user" : NULL
    "count" : 3
    "recurring" : true
  }
  4 : {
    "alarm_id" : "ALM-A001-010"
    "asset_id" : "A001"
    "alarm_name" : "High Discharge Pressure"
    "severity" : "high"
    "status" : "cleared"
    "start_time" : "2026-10-12T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Discharge pressure exceeding setpoint"
    "end_time" : "2026-10-12T20:32:32.777685Z"
    "ack_time" : "2026-10-12T19:02:32.777685Z"
    "ack_user" : "operator-1"
    "count" : 3
```

```
    "recurring" : true
}
5 : {
    "alarm_id" : "ALM-A001-009"
    "asset_id" : "A001"
    "alarm_name" : "Motor Trip"
    "severity" : "medium"
    "status" : "active"
    "start_time" : "2026-10-06T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "safety"
    "description" : "Motor protection trip activated"
    "end_time" : NULL
    "ack_time" : NULL
    "ack_user" : NULL
    "count" : 3
    "recurring" : true
}
6 : {
    "alarm_id" : "ALM-A001-008"
    "asset_id" : "A001"
    "alarm_name" : "Flow Rate Low"
    "severity" : "low"
    "status" : "acknowledged"
    "start_time" : "2026-09-30T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Flow rate below minimum threshold"
    "end_time" : NULL
    "ack_time" : "2026-09-30T19:02:32.777685Z"
    "ack_user" : "operator-1"
    "count" : 3
    "recurring" : true
}
7 : {
    "alarm_id" : "ALM-A001-007"
    "asset_id" : "A001"
    "alarm_name" : "Temperature High"
    "severity" : "critical"
    "status" : "cleared"
    "start_time" : "2026-09-24T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Operating temperature above safe limit"
    "end_time" : "2026-09-24T20:32:32.777685Z"
    "ack_time" : NULL
    "ack_user" : NULL
    "count" : 3
    "recurring" : true
}
8 : {
    "alarm_id" : "ALM-A001-006"
```

```
    "asset_id" : "A001"
    "alarm_name" : "Vibration Alarm"
    "severity" : "high"
    "status" : "active"
    "start_time" : "2026-09-18T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "device"
    "description" : "Excessive vibration detected"
    "end_time" : NULL
    "ack_time" : "2026-09-18T19:02:32.777685Z"
    "ack_user" : "operator-1"
    "count" : 3
    "recurring" : true
  }
▼ 9 : {
    "alarm_id" : "ALM-A001-005"
    "asset_id" : "A001"
    "alarm_name" : "High Discharge Pressure"
    "severity" : "medium"
    "status" : "acknowledged"
    "start_time" : "2026-09-12T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Discharge pressure exceeding setpoint"
    "end_time" : NULL
    "ack_time" : NULL
    "ack_user" : NULL
    "count" : 3
    "recurring" : true
  }
▼ 10 : {
    "alarm_id" : "ALM-A001-004"
    "asset_id" : "A001"
    "alarm_name" : "Motor Trip"
    "severity" : "low"
    "status" : "cleared"
    "start_time" : "2026-09-06T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "safety"
    "description" : "Motor protection trip activated"
    "end_time" : "2026-09-06T20:32:32.777685Z"
    "ack_time" : "2026-09-06T19:02:32.777685Z"
    "ack_user" : "operator-1"
    "count" : 3
    "recurring" : true
  }
▼ 11 : {
    "alarm_id" : "ALM-A001-003"
    "asset_id" : "A001"
    "alarm_name" : "Flow Rate Low"
    "severity" : "critical"
    "status" : "active"
```

```
    "start_time" : "2026-08-31T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Flow rate below minimum threshold"
    "end_time" : NULL
    "ack_time" : NULL
    "ack_user" : NULL
    "count" : 3
    "recurring" : true
  }
  12 : {
    "alarm_id" : "ALM-A001-002"
    "asset_id" : "A001"
    "alarm_name" : "Temperature High"
    "severity" : "high"
    "status" : "acknowledged"
    "start_time" : "2026-08-25T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Operating temperature above safe limit"
    "end_time" : NULL
    "ack_time" : "2026-08-25T19:02:32.777685Z"
    "ack_user" : "operator-1"
    "count" : 3
    "recurring" : true
  }
  13 : {
    "alarm_id" : "ALM-A001-001"
    "asset_id" : "A001"
    "alarm_name" : "Vibration Alarm"
    "severity" : "medium"
    "status" : "cleared"
    "start_time" : "2026-08-19T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "device"
    "description" : "Excessive vibration detected"
    "end_time" : "2026-08-19T20:32:32.777685Z"
    "ack_time" : NULL
    "ack_user" : NULL
    "count" : 3
    "recurring" : true
  }
  14 : {
    "alarm_id" : "ALM-A001-000"
    "asset_id" : "A001"
    "alarm_name" : "High Discharge Pressure"
    "severity" : "low"
    "status" : "active"
    "start_time" : "2026-08-13T18:32:32.777685Z"
    "asset_name" : "Boiler Feed Pump 101"
    "alarm_type" : "process"
    "description" : "Discharge pressure exceeding setpoint"
```

```
      "end_time" : NULL
      "ack_time" : "2026-08-13T19:02:32.777685Z"
      "ack_user" : "operator-1"
      "count" : 3
      "recurring" : true
    }
  ]
}
```


Operator Recommendations


```
{
  "alarm_id" : "ALM-A001-003"
  "immediate_actions" : [
    0 : "Verify the Flow Rate Low indication against current instrument readings
    1 : "Follow the approved site procedure and operating limits before taking a
  ]
  "preventive_measures" : [
    0 : "Schedule preventive maintenance"
    1 : "Review operating procedures"
  ]
  "related_assets" : [
    0 : "A002"
    1 : "A003"
  ]
  "documentation_references" : [
    0 : "SOP-101"
    1 : "Maintenance-Manual-Ch3"
  ]
  "trace_id" : "acd895c4-93b9-44c2-9249-fb6537651074"
  "x_client_id" : "alarm-copilot"
}
```


Procedure evidence (4)

pump_operating_procedure.txt · 56% match

OPERATING PROCEDURE FOR BOILER FEED PUMP SYSTEMS ABB Industrial Automation Guide - Documer Feed Pumps are critical equipment in thermal power plants and refineries. These pumps are responsibl supply to boilers under high pressure and elevated temperatures. Proper operation and monitoring are operation. 2. NORMAL OPERATING CONDITIONS The Boiler Feed Pump (BFP) operates under the followi Pressure: 250-280 bar - Suction Pressure: 2-5 bar (depending on feed water heater configuration) - Opera Rate: 100-150 m^3/h (asset dependent) - Vibration Level: < 4.5 mm/s RMS - Motor Current: 85-110 A 2.1 ST line isolation valve is fully open 2. Open drain valves to release trapped air 3. Verify discharge line back p using control system 5. Monitor pressure rise on discharge gauge 6. Check for leakage at gland seals and stabilization 8. Record baseline vibration reading 2.2 NORMAL OPERATION - Monitor discharge pressure vibration or noise - Verify motor current consumption is normal - Maintain operating log with 4-hour inte applicable - Monitor bearing temperature 3. ALARM MANAGEMENT 3.1 High Discharge Pressure Alarm (2 pressure exceeding safe operating limits. CAUSES: - Downstream blockage or valve closure - Increased s failure - Check valve stuck closed - Control valve calibration error IMMEDIATE ACTIONS: 1. Reduce pump 2. Check discharge line for blockages 3. Verify downstream pressure relief valve functionality 4. Open dis pressure remains high, reduce suction valve opening to 75% 6. Contact maintenance if unable to norma - Check downstream equipment for operational status - Review recent modifications to discharge line - V are open - Test pressure relief valve setpoint 3.2 Vibration Alarm (> 5 mm/s) Excessive vibration indicates Cavitation at pump inlet (common cause) - Bearing wear or misalignment - Impeller wear or damage - F

line blockage - Cavitation from low suction pressure IMMEDIATE ACTIONS: 1. Check suction pressure gau Inspect for unusual noise patterns 3. Check feed water heater outlet temperature 4. Verify suction strain speed by 10-20% to test 6. If vibration persists, prepare for shutdown INVESTIGATION: - Review recent su water heater performance - Verify suction line is fully open - Inspect suction strainer differential pressure Temperature High Alarm (185°C) Operating temperature above safe limit indicates thermal stress. CAUSI Low feed water temperature from preheater - High discharge pressure requiring throttling - Bearing cool induced friction heating IMMEDIATE ACTIONS:

Metadata


```
{
  "source" : "pump_operating_procedure.txt"
  "doc_type" : "procedure"
  "chunk_index" : 0
  "total_chunks" : 2
  "start_word" : 0
  "end_word" : 500
  "ingestion_date" : "2026-11-11T14:08:22.606899+00:00"
}
```


motor_troubleshooting_guide.txt · 46% match

PREVENTIVE MAINTENANCE SCHEDULE DAILY - Visual inspection for leaks or damage - Listen for unusua temperature with touch - Record vibration level if instrumented WEEKLY - Measure bearing temperature around motor - Inspect for dust or debris accumulation MONTHLY - Measure insulation resistance (mego if water-cooled - Check for any visible cracking or damage - Review operational hours counter QUARTER spectrum analysis - Inspect motor mounting bolts - Check bearing lubrication - Measure winding tempe service inspection - Bearing replacement if > 40,000 hours - Insulation system testing - Shaft alignment c ALARM RESPONSE PROCEDURES Motor Overload Alarm (Thermal Protection) - Indicates motor current c Cause: High pump discharge pressure, cavitation, or mechanical resistance - Action: Reduce pump spee suction - If alarm clears within 2 minutes: Monitor closely - If alarm persists: Reduce load further or initia Alarm - Occurs when three-phase motor receives unbalanced current - Cause: Electrical supply issue or r electrical maintenance immediately - Do not continue operation if imbalance > 5% Motor Bearing Temp bearing temperature > 85°C - Common cause: Pump cavitation, high vibration - Action: Reduce pump sp pressure and vibration - If temperature does not drop within 10 minutes: Shutdown 6. MOTOR SELECTIO selecting replacement motors for high-reliability systems: - Ensure motor is sized for actual pump duty (efficiency motor (IE3 or better) - Consider thermal class F for thermal margin - Request thermal protecti bearing design is appropriate for vibration environment - Check alignment and coupling quality 7. MAIN Record for every motor service event: - Date and time - Service performed (inspection, lubrication, repai after - Bearing condition assessment - Any replacement parts used - Operator and technician names - Ne recommendations noted --- Revision: 3.2 | Issue Date: 2025-11-15 | Next Review: 2026-11-15

Metadata


```
{
  "source" : "motor_troubleshooting_guide.txt"
  "doc_type" : "procedure"
  "chunk_index" : 1
  "total_chunks" : 2
  "start_word" : 400
  "end_word" : 777
  "ingestion_date" : "2026-11-11T14:08:22.252405+00:00"
}
```


pump_operating_procedure.txt · 37% match

motor_troubleshooting_guide.txt · 32% match

MCP execution trace (3)

Complete · `search_assets` · 688 ms

Request arguments


```
{
  "query" : "Boiler Feed Pump 101"
  "limit" : 10
}
```


Response


```
{
  "total" : 1
  "results" : [
    0 : {
      "asset_id" : "A0
      "asset_name" : '
      "asset_type" : '
      "unit" : "Unit 1
      "location" : "Ea
      "criticality" :
    }
  ]
}
```


Complete · `get_alarms` · 26 ms

Request arguments


```
{
  "asset_id" : "A001"
  "page" : 1
  "page_size" : 200
}
```


Response


```
{
  "total" : 15
  "page" : 1
  "page_size" : 200
  "total_pages" : 1
  "data" : [...]
}
```


Complete · `operator_recommendations` · 19 ms

Request arguments


```
{
  "alarm_id" : "ALM-A001-003"
}
```


Response


```
{
  "alarm_id" : "ALM-A001
  "immediate_actions'
  "preventive_measure
  "related_assets" : [
  "documentation_refe
  "trace_id" :
  "acd895c4-93b9-44c2-92
  "x_client_id" : "alarm
}
```


Investigation ID: `acd895c4-93b9-44c2-9249-fb6537651074`

AI-generated guidance can be incomplete. Verify safety-critical actions against approved site procedures

Describe an alarm or asset to investigate…